DraftRun Pipeline TO BE 2.17.4.6.1.3.5

Target design for the DraftRun provider-input treatment track: `2.17.4.6.1.3.5` through `2.17.4.6.1.3.10`.

This document is the target architecture for fixing oversized provider context across DraftRun. It is not current behavior. The AS IS baseline remains `docs/architecture/DRAFT_RUN_PIPELINE_AS_IS.md`.

PDF quick-view copy: `docs/architecture/DRAFT_RUN_PIPELINE_TO_BE_2_17_4_6_1_3_5.pdf`.

Regenerate PDF:

```
python scripts/generate-draft-run-pipeline-pdf.py `
  --source docs/architecture/DRAFT_RUN_PIPELINE_TO_BE_2_17_4_6_1_3_5.md `
  --output docs/architecture/DRAFT_RUN_PIPELINE_TO_BE_2_17_4_6_1_3_5.pdf
```

1. Target change summary

Change marker	Meaning
NEW in 2.17.4.6.1.3.5	Provider-input audit and hard budget enforcement for every provider-heavy DraftRun child `AiRun`.
NEW in 2.17.4.6.1.3.6	Deterministic `DraftRunContextAccessService` and typed provider-input dossier factories.
CHANGED vs AS IS	Provider-heavy prompt builders no longer receive full `rulePack`, `materialPlan`, candidate pools, validation reports, or source ledgers directly.
UNCHANGED	DraftRun step order, prompt goals, model roles, public HTTP API, SQLite schema, and UI layout.
NOT this track	Broad prompt-quality rewrite, model replacement policy, autonomous agent loop, or mandatory external MCP server.

Main target:

Provider-heavy operations must receive a compact operation-specific dossier, not a large copy of the full DraftRun working set.

The rich artifacts still exist in parent DraftRun storage and diagnostics. The provider boundary becomes a deliberate projection layer:

`DraftRun artifacts -> ContextAccess -> DossierFactory -> BudgetGate -> PromptBuilder -> Provider`

2. Why this track exists

A live audit over six DraftRuns showed that the long `materialPlan` call was only one symptom. Other steps already send equal or larger provider inputs:

Operation	Largest observed input	Approx tokens	Direct provider-input budget
`pairwiseRanking`	626449 chars	156613	missing
`materialPlan`	346264 chars	86566	missing
`alternativeAngleCandidate`	328412 chars	82103	missing
`draftCandidate`	328276 chars	82069	missing
`strategy`	323040 chars	80760	missing
`alternativeAngleRoute`	320833 chars	80209	missing

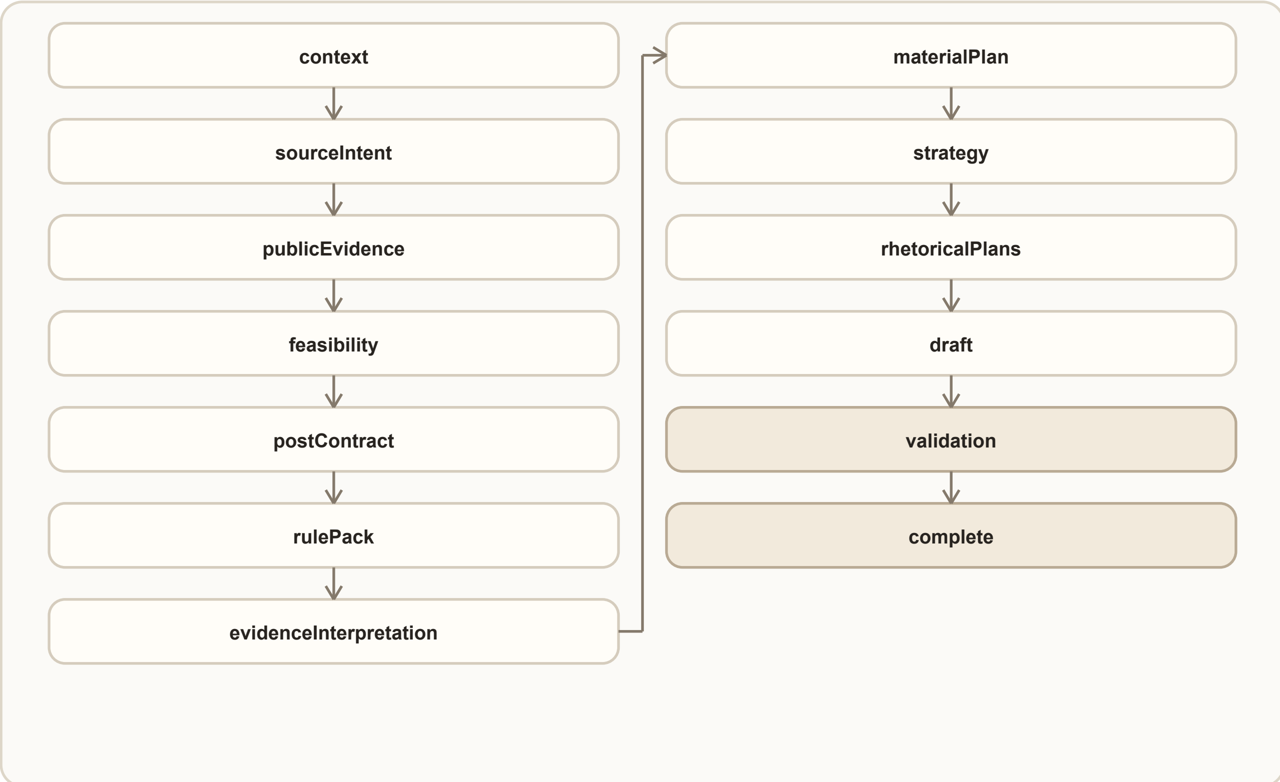
Operation	Largest observed input	Approx tokens	Direct provider-input budget
`llmValidation`	255935 chars	63984	missing
`rhetoricalPlans`	168461 chars	42116	missing

The current budget layer exists but is only enforced for representative operations such as `editorialCritique` and `directedRevision`. Some traces contain `payloadBudget` only because a previous artifact includes another operation's budget metadata. That is not the same as a direct budget on the current provider call.

The risk is not only cost or latency. Oversized context also weakens model focus, raises timeout and schema-failure risk, and makes quality failures harder to explain.

3. Target step order

No persisted DraftRun step is added. The existing order stays intact:



CHANGED vs AS IS: provider-heavy sub-operations inside those steps cross a new input boundary before prompt messages are built.

4. Target role-aware execution map

Operation family	Active role	Target provider input
`evidenceInterpretation`	`strategy`	Evidence interpretation dossier with accepted evidence, claim handles, contract constraints, and relevant rules.
`materialPlan`, `strategy`, `rhetoricalPlans`	`strategy`	Planning dossier with compact contract, interpreted evidence, selected rule ids, usable evidence handles, and planning constraints.
`draftCandidate`, `alternativeAngleCandidate`	`writer`	Writer dossier with one direction, material decisions, evidence handles, allowed claims, forbidden moves, and size/style constraints.

Operation family	Active role	Target provider input
`alternativeAngleRoute`	`anotherAngle`	Alternative-angle dossier with selected candidate summaries, critique signals, contract invariants, and rejected-move constraints.
`llmValidation`, `editorialCritique`	`review` / `critic`	Review dossier with one candidate, compact deterministic findings, relevant claims/rules, and evidence interpretation summaries.
`pairwiseRanking`	`review`	Ranking dossier with compact candidates, scorecard dimensions, validation summaries, material evidence handles, and final selection constraints.
`directedRevision`, final repair	`writer`	Revision dossier with selected candidate, repair goals, anti-regression constraints, and referenced evidence/rule handles.
`finalQualityGateReview`	`finalGate`	Final-quality dossier with final prose, final quality contract, open issue lifecycle, attribution summary, and repair history.

5. Controlled architecture

5.1 Rich working set stays rich

UNCHANGED: parent DraftRun artifacts keep full diagnostic data:

- `SourceLedger`;
- `PublicEvidenceBatch`;
- `EvidenceSynthesis`;
- `RuleRegistrySnapshot`;
- `EvidenceInterpretation`;
- `MaterialPlan`;
- `DraftCandidates`;
- validation reports;
- ranking/revision/final quality traces.

Full data is for storage, diagnostics, replay, and human inspection. It is not the default provider input.

5.2 Context access service

NEW in 2.17.4.6.1.3.6: add a provider-free `DraftRunContextAccessService` under `backend.app.drafting.application`.

It exposes deterministic reads such as:

- get compact post contract;
- get relevant claim handles;
- get accepted evidence summaries;
- get rule summaries by scope/severity;
- get candidate summaries;
- get validation issue summaries;
- get final quality issue lifecycle;

- resolve a handle back to the full artifact for trace/debug.

The service does not call providers and does not write DraftRun state.

5.3 Dossier factories

NEW in 2.17.4.6.1.3.6: add role-owned dossier factories:

- `PlanningDossierFactory`;
- `WriterDossierFactory`;
- `ReviewDossierFactory`;
- `RankingDossierFactory`;
- `RevisionDossierFactory`;
- `FinalQualityDossierFactory`.

Each factory returns a typed DTO or trace-safe payload with:

- `profileId`;
- `operationId`;
- `mustHave`;
- `shouldHave`;
- `diagnosticOnly`;
- `neverSendToProvider`;
- compact data sent to the provider;
- handles for omitted full artifacts;
- sent/trimmed/suppressed counts;
- quality-risk notes.

5.4 Budget gate

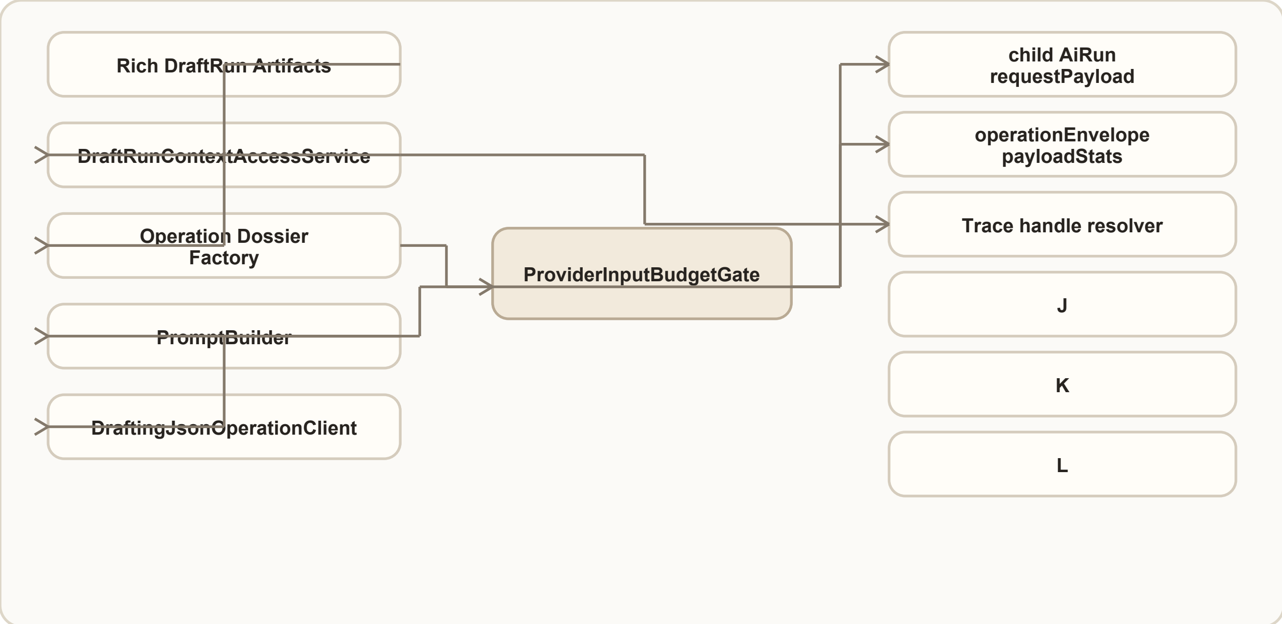
NEW in 2.17.4.6.1.3.5: every provider-heavy operation must pass a direct `ProviderInputBudgetGate` before `build_*_messages(...)`.

The gate records:

- `payloadBudget.profileId`;
- execution mode;
- max prompt chars and approximate token budget;
- prompt char estimate;
- approximate token estimate;
- sent counts;
- trimmed counts;
- suppressed fields;
- `contextOverBudget` or `payloadTooLarge` incident when applicable.

The gate must inspect the current call, not nested historical artifacts.

6. Target artifact flow



Trace principle:

- provider request stores the compact dossier and budget metadata;
- parent artifact stores the rich data;
- trace can resolve handles back to rich artifacts for debugging;
- no prompt builder receives full artifacts unless the operation has an explicit

temporary debt entry and architecture smoke allows it.

7. Step-by-step TO BE flow

7.1 Provider operation runtime guard

NEW in 2.17.4.6.1.3.4: extend the current queue/staleness repair into a provider operation runtime guard.

It records:

- current provider operation id;
- operation start time;
- direct prompt char estimate;
- selected model;
- provider wait time;
- slow-but-healthy status;
- stale/timeout reason if the operation exceeds budget.

This prevents a slow model call from looking like an unexplained worker hang.

7.2 Provider-input audit and enforcement

NEW in 2.17.4.6.1.3.5: add a repeatable audit over stored child `AiRun` records.

The audit must identify:

- provider-heavy calls missing direct `payloadBudget`;
- calls above operation caps;
- prompt builders that pass full `rulePack`, `sourceLedger`, `materialPlan`,

candidate pool, validation report, or final quality trace;

- false-positive nested budget metadata.

Architecture smoke should fail new provider-heavy code that bypasses the budget gate.

7.3 Planning dossier migration

IMPLEMENTED in 2.17.4.6.1.3.7: migrated runtime provider calls:

- `materialPlan`;
- `strategy`;
- `rhetoricalPlans`.

These operations now receive operation-specific planning dossiers assembled from persisted DraftRun artifacts through `DraftRunContextAccessService`:

- compact post contract;
- interpreted evidence summaries;
- selected claim/evidence/rule handles;
- material-planning constraints;
- rejected-use summaries.

They should not receive full `rulePack`, full `SourceLedger`, or embedded previous operation traces.

Each primary, repair, and backup attempt crosses `ProviderInputBudgetGate` after the dossier is assembled. Child `AiRun.requestPayload` records the actual `providerInput`, `providerDossier.runtimeMigrated=true`, direct budget proof, and attempt metadata. Rich artifacts remain available to deterministic fallback and parent trace, but prompt builders no longer accept them. A `BLOCKED` planning dossier skips the provider and uses the existing safe fallback; `DEGRADED` is valid only when all `mustHave` inputs remain present.

Live proof: `c2303e05-e7d0-4cad-a3f9-6ea26fc1a3ed` completed the full pipeline with all three planning inputs below their standard caps, zero forbidden-field or unresolved-handle findings, sufficient evidence coverage, and no open editorial issues. The comparison against baseline `e874fd2b-cfa0-4b6a-815d-c0cf6d9763d2` is committed under `docs/evidence/draft-runs/<baseline-id>/`.

7.4 Writer and alternative-angle dossier migration

IMPLEMENTED in 2.17.4.6.1.3.8: migrated runtime provider calls:

- `draftCandidate`;
- `alternativeAngleRoute`;
- `alternativeAngleCandidate`.

`WriterDossierFactory` now builds operation-specific `draftCandidate` and `alternativeAngleCandidate` projections. `AlternativeAngleDossierFactory` builds the route projection from persisted candidate summaries, critique signals, validation issues, rejected moves, contract invariants, and supporting handles. The accepted route is persisted before challenger dossier construction. Every attempt crosses `ProviderInputBudgetGate` and records `providerDossier.runtimeMigrated=true`, direct `providerInput`, `payloadBudget`, `inputStats`, and `payloadStats`.

The implemented standard-mode projections are intentionally bounded before prompt construction: primary writer calls retain four curated evidence items, two claim handles and five rules; the challenger retains the accepted route, four evidence items, one claim handle, four rules and one critique summary. Route selection retains three candidate summaries, three critique summaries, three validation issues, two rejected moves, two evidence items and two rules. These limits keep `draftCandidate` and `alternativeAngleCandidate` within 24000 characters and `alternativeAngleRoute` within 22000 while preserving every semantic `mustHave` field.

Full planning stacks, candidate pools, validation traces, old envelopes/budgets, and role `ContextPacks` remain in parent DraftRun storage and are not provider input.

Live proof:

- `72b3a2df-c8be-41a8-b2c0-e74e5e502cd0` completed the full pipeline after the

final compaction pass;

- `draftCandidate` recorded direct budget proof at 16256-16533 chars against the

24000 cap;

- `alternativeAngleRoute` recorded 18386 chars against the 22000 cap;
- `alternativeAngleCandidate` recorded 16534 chars against the 24000 cap;
- replay reported zero unresolved handles and zero forbidden-field violations.

A later broader live context in run `7bf3...` exposed data-dependent drift in `alternativeAngleRoute`: `34589/22000`. The migration remains structurally correct, but its projection is not reliably bounded. Repair is explicit follow-up `2.17.4.6.1.3.9.1`; it must preserve the semantic route contract while enforcing the cap before the tool-mediated pilot.

Slice `2.17.4.6.1.3.9.1` implements that repair. Route selection now applies a dedicated deterministic compactor with equal per-candidate fields and budgets, severity-prioritized critique/validation signals, bounded contract/evidence/rules, and a reserved repair-context budget. The same gate sees dossier plus repair context, then `ProviderMessageBudgetGuard` verifies the final serialized messages. Live run `92532cb9-e83b-4bb1-ab2b-7d7a46d279b5` records `14371` provider-input characters and `16321/22000` actual message characters, zero route budget incidents, no missing required input, a generated challenger, and a challenger-derived final winner. This closes the `3.9.1` follow-up without changing route output semantics.

The final proof also showed why Slice `3.9` remains necessary: the delivered text itself stayed below `hardMaxChars`, but `qualityFidelity.issueLifecycle` still counted open critical findings from intermediate or losing validation reports. That is not a writer-dossier regression; it is review/ranking/final-gate lifecycle debt. The comparison is stored in `docs/evidence/draft-runs/e874fd2b-cfa0-4b6a-815d-c0cf6d9763d2/COMPARISON_2_17_4_6_1_3_8.md`.

7.5 Review, ranking, and final gate dossier migration

NEW in 2.17.4.6.1.3.9: migrate:

- `llmValidation`;
- `pairwiseRanking`;
- `directedRevision`, including final repair;
- `finalQualityGateReview`;
- remaining final repair review calls.

Ranking must compare compact candidate summaries and score dimensions. It should not receive full candidate pool plus full material plan plus full validation report when only candidate ids, short body excerpts, issue summaries, and scorecard dimensions are needed.

`qualityFidelity.issueLifecycle` must also stop treating all intermediate and losing candidate reports as open final-draft quality blockers. The lifecycle should make the final selected or repaired candidate explicit, keep non-final findings visible as diagnostic context, and block trusted quality only for unresolved issues that still apply to the delivered final text.

Implementation status (2026-07-12): runtime wiring, direct attempt budgets, terminal replay, persisted-candidate recovery, and candidate-scoped lifecycle are implemented. `72b3...` replay now has `openCriticalCount=0` and `openWarningCount=1`; losing-candidate findings remain diagnostic. Exact-payload replay of the final compactor over `d06...` keeps all four operation families inside their caps with zero forbidden/unresolved handles.

Section 7.5 is implemented and live-accepted. Fresh control run `7bf3...` records accepted maximums `llmValidation=16628/22000`, `pairwiseRanking=20534/22000`, `directedRevision=20157/24000`, and `finalQualityGateReview=18015/22000`, with all 16 target attempts directly budgeted. Evidence fidelity is `sufficient`, editorial status is `publishable`, open critical/warning counts are 0/0, and final repair is reviewed and accepted against the actually delivered candidate. Evidence: `docs/evidence/draft-runs/e874fd2b-cfa0-4b6a-815d-c0cf6d9763d2/COMPARISON_2_17_4_6_1_3_9.md`.

7.6 Tool-mediated context access pilot

NEW in 2.17.4.6.1.3.10: pilot a tool-mediated context access path for one operation after the deterministic context service exists.

The tool or MCP adapter is optional. It must call deterministic context access methods and must not expose raw full DraftRun JSON to the model.

8. Trace contract

Every migrated provider-heavy child `AiRun.requestPayload` must show:

```
{
  "draftRunStep": "materialPlan",
  "operationId": "materialPlan",
  "providerInput": {
    "dossierId": "planningDossier:standard",
    "sent": {},
    "handles": {}
  },
  "payloadBudget": {
    "profileId": "materialPlan",
    "executionMode": "standard",
    "promptCharEstimate": 24000,
    "approxTokenEstimate": 6000,
    "sentCounts": {},
    "trimmedCounts": {},
    "suppressedFields": [],
    "qualityRisk": "none"
  }
}
```

Every parent step artifact may keep the full source data. `/ai-runs?runId=...` should show both:

- what the provider actually received;
- which full artifacts can be inspected through handles.

9. Acceptance criteria

The provider-input treatment track is successful when:

- every provider-heavy operation has a direct budget profile or an explicit
temporary debt entry;
- prompt builders no longer receive full `rulePack`, `SourceLedger`,
`materialPlan`, candidate pool, validation report, or final quality trace by default;
- `pairwiseRanking`, `draftCandidate`, `alternativeAngle*`, `strategy`,
`materialPlan`, `llmValidation`, and `rhetoricalPlans` all show lower, operation-specific prompt estimates in child `AiRun.requestPayload`;
- full parent artifacts remain available for diagnostics;
- live DraftRun proof shows no loss of required claims/rules/evidence handles;
- quality/fidelity diagnostics remain at least as strict as before;
- architecture smoke fails new provider-heavy operations that bypass the budget
gate.

10. Implementation boundaries

Allowed:

- typed provider-input DTOs;
- deterministic context access service;
- operation-specific dossier factories;
- budget gate enforcement;
- trace-safe handle references;
- smoke/audit checks;
- replay and live proof diagnostics.

Not allowed:

- public HTTP contract changes;
- SQLite schema migration;
- changing DraftRun step order;
- changing model selection policy;
- broad prompt-quality rewrite;
- replacing OpenRouter adapter;
- hiding quality warnings by trimming context.

11. Explicit non-goals

This track does not promise that every final draft becomes publishable. It fixes the provider-input architecture so later quality work is not fighting uncontrolled context bloat.

It also does not make MCP mandatory. Tool-mediated context access is a later pilot after the deterministic context access service exists.

12. Maintenance rule

When any slice in this track is implemented:

- update this TO BE if the target changes;
- update `docs/architecture/DRAFT_RUN_PIPELINE_AS_IS.md` only when runtime behavior actually changes;
- update `docs/architecture/BACKEND_ARCHITECTURE_TARGET.md` and `backend/app/drafting/README.md` when new owners are introduced;
- regenerate this PDF after changing the TO BE;
- run `npm run test:architecture`, roadmap `render/export/check`, and `git diff --check`.